

Do domestic herbivores retard *Polylepis australis* Bitt. woodland recovery in the mountains of Córdoba, Argentina?

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Abstract

Herbivores are often an important factor hindering the recovery of woodlands. Browsing generally retards seedling and juvenile tree growth, but the exact impact of herbivores is still controversial. The high mountain areas of central Argentina consist of a mosaic of *Polylepis australis* Bitt. woodlands and different grassland types which are grazed by large domestic herbivores. To contribute to the management of these ecosystems, we performed an observational and an experimental study. In the first, we measured *P. australis* browsing intensity and size structure in areas with sparse woodland cover under two treatments. One treatment consisted of seven areas with high stocking rates, and the other consisted of five areas where livestock were reduced to low or moderate stocking rates 6–7 years prior to our measurements. In the experiment, we planted *P. australis* seedlings in a heavy grazed and an ungrazed area and measured growth and survival during 6 years. Our observational study showed that livestock browse heavily on *P. australis*, but a reduction of livestock densities reduced browsing rates and changed population size structure. The experimental results confirmed the high browsing rates, and showed that browsing negatively influences survival and height-growth. Our data suggests that heavy grazing by domestic herbivores retard *P. australis* woodland recovery, but a reduction in stocking rates promote changes that can lead to an increase in *P. australis* density. An adequate livestock management is therefore very important in the mountains of Córdoba.

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1. Introduction

Herbivores can affect woodland structure (McInnes et al., 1992; Homolka and Herolodová, 2003) and even preserve other vegetation types such as grasslands or shrublands in sites with potential to

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develop into woodlands (Anderson, 1981; Hope et al., 1996; Vera, 2000; Rao et al., 2003). While the initial opening of woodlands and transformation into other physiognomic types is generally driven by fire, browsing may maintain these physiognomies (Anderson, 1981; Kramer et al., 2003) as it might affect the development of seedlings and saplings (Danell et al., 2003), as well as the maturation of individuals (McNaughton and Sabuni, 1988; Motta, 2003). There is, however, great variability in the effects of browsing on plant population dynamics, depending on factors such as the susceptibility of the woody species involved (e.g. Sun et al., 1997; Garin et al., 2000; Motta, 2003), the herbivore behaviour, which in turn might be influenced by habitat characteristics or the presence of predators (Garin et al., 2000; Palmer and Truscott, 2003; White et al., 2003; Pietrzykowski et al., 2003) and mortality due to factors other than browsing (Senn and Sutter, 2003; Palmer and Truscott, 2003).

The maintenance of grasslands in sites potentially dominated by woody vegetation might be desirable both from the productive and conservation standpoint, and in some ecosystems grazing management is oriented towards that goal (Dolman and Sutherland, 1991; Hope et al., 1996; Dolek and Geyer, 2002; Vera, 2000). On the other hand, the restoration of woodlands or forests might be essential to recover ecosystem services provided by these vegetation types, including hydrological regime regulation, prevention of soil erosion and conservation of biodiversity (Spies, 1998; Senn and Sutter, 2003; Rao et al., 2003; Zak et al., 2004). In these cases, the complete exclusion of large herbivores will often secure rapid growth of juvenile trees previously affected by browsing, but it could also result in losses of biodiversity (Mitchell and Kirby, 1990), increased fire hazard due to fuel accumulation (Belsky and Blumenthal, 1997) and reduced seedling establishment due to competition (Romagosa and Robison, 2003). Consequently, the control of grazing rather than its complete removal is now generally considered as a more desirable management option (Garin et al., 2000). Therefore, it is necessary to understand the interactions between herbivores and vegetation, both if the management objective is to maintain woodlands, or to maintain grasslands (Hunter, 1990; Garin et al., 2000; Weisberg and Bugmann, 2003).

At the upper mountain and subalpine belts, woodlands are usually more vulnerable to disturbance than lowland forests (Ammer, 1996; Motta, 2003). In the higher parts of the mountains of Córdoba (central Argentina), *Polylepis australis* Bitt. closed woodlands and open woodlands/shrublands occupy about 12% of the area, but there is evidence that these communities covered substantially larger areas in the past (Cabido and Acosta, 1985; Renison et al., 2002, 2004, in press; Cingolani et al., 2004). In 1997, part of the area (26,000 ha) was expropriated to create the Quebrada del Condorito National Park, while the private lands surrounding the Park, although not expropriated, were declared National Reserve (12,000 ha) and Provincial Reserve (117,000 ha). Following the “The Nature Conservancy” methodology (TNC, 2002), *P. australis* woodlands, as well as the mosaic of different types of grasslands, were designated as two important conservation objects for the Conservation Unit formed by the three protected areas (APN, 2004). To conserve the mosaic of grasslands, domestic livestock have been maintained in some areas of the National Park to prevent the excessive dominance of tussock grasslands at the expense of grazing lawns, which in the past were probably maintained by the locally extinct *Lama guanicoe* herds (Díaz et al., 1994; Pucheta et al., 1998; Cingolani et al., 2003). However, there is some evidence that livestock may prevent long-term recovery of *P. australis* woodlands (Renison et al., in press). This would imply a conflict between conservation objects, because livestock, which at present are indispensable for maintaining the mosaic that maximises grassland biodiversity, could be hampering the recover of woodlands.

To develop a suitable management strategy capable of meeting both conservation purposes, it is necessary to understand the effects of livestock on *P. australis* woodland development. To achieve this objective, we (1) measured *P. australis* browsing intensity and size structure in areas with high grazing pressure and compared them with areas where grazing pressure was reduced 6–7 years prior, all areas having low present *P. australis* cover and (2) performed a 6-year field experiment planting *P. australis* seedlings in a heavy grazed and an ungrazed area.

2. Methods

2.1. Study area

The study was carried out in the upper portion of Sierras Grandes, a mountain chain located in Córdoba, central Argentina (1700–2800 m a.s.l., its central point at 31°34'S, 64°50'W). Because these mountains are 1000 m higher than the surrounding lands, they constitute a biogeographical island (Cabido et al., 1998) with 41 endemic plant and animal taxa (Cabido et al., 2003). Mean temperature of the coldest and warmest months are 5.0 and 11.4 °C, respectively, and there is no frost-free period. Mean annual precipitation is 920 mm (1992–2000), with most rainfall concentrated in the warmer months, between October and April (Cabido, 1985; Colladon, 2000). The main economic activity is livestock rearing (primarily cattle), which began early in the 17th century and completely replaced large native herbivores (*L. guanicoe*, and probably *Rhea americana*) by the beginning of the 20th century (Díaz et al., 1994).

The area comprises different landscape units, including valley bottoms and ravines, plateaus with different degrees of dissection, rocky hilly uplands and steep escarpments (Cabido et al., 1987). Most of these units (with the exception of plateaus with low dissection) are rough, with abundant rocky outcrops, steep slopes and high topographic variability at short distances (Cingolani et al., 2004). Vegetation consists of a mosaic of tussock grasslands, grazing lawns, granite outcrops, *P. australis* woodlands, and eroded areas with exposed rock surfaces (Cabido, 1985; Cabido and Acosta, 1985; Funes and Cabido, 1995; Cingolani et al., 2003, 2004). Due to its intrinsic fragility and three centuries of domestic grazing and anthropogenic fires, the mountain range has now serious problems of erosion and woodland degradation (Cabido and Acosta, 1985; Renison et al., 2002, 2004, in press; Cingolani et al., 2003, 2004). At present, woodlands with a closed canopy occupy only 2.5% of the area, while open woodlands mixed with grasslands and rocky outcrops occupy another 9.4% (Cingolani et al., 2004). There is also a third type of woodlands, consisting in relatively small patches of sparsely distributed *P. australis* individuals (canopy cover less than 10%). These sparse woodlands,

although very inconspicuous, are present with relatively high frequency in most landscape units.

There are evidences that closed woodlands occupied larger areas in the past, having *P. australis* density decreased due to the action of anthropogenic fires and grazing (Cabido and Acosta, 1985; Renison et al., 2002, 2004, in press). Thus, it is expected that open or sparse woodlands, if protected from grazing and fire, would develop into closed woodlands. In this study, we analysed only sparse woodlands (less than 10% canopy cover), since due to their ubiquity, they could be important for developing woodland recovery management strategies.

2.2. Observational study

2.2.1. Field sampling

We selected 12 areas (0.25–0.6 ha) of sparse *Polylepis* woodlands distributed in the central and northern portion of the Sierras Grandes range. Areas were as internally homogeneous as possible given the high heterogeneity of these mountains; their size and limits being defined by the presence of *P. australis* individuals, together with physiognomic and topographic features. Seven of those areas were under high grazing intensities and five were under low to moderate present grazing intensity. Areas under high grazing intensity were located in different private lands and represent the usual situation in privately owned ranches. Four of the five areas with low/moderate grazing intensity were located in different paddocks of the National Park where livestock density was reduced in 1998, and one in a privately owned valley where livestock was reduced due to management reasons, in 1999 (owner pers. comm.). Areas belonging to different treatments (high grazing intensity and reduced grazing intensity) were as interspread as possible, given the restrictions imposed by the availability of suitable areas with reduced grazing. Areas belonging to the same treatment were at least 2 km apart from each other. All areas were primarily grazed by cattle, although horses were present at low stocking rates, and occasionally sheep and goats were also observed. Fires have not occurred in the selected areas in the last 8 years (National Parks personnel, pers. comm. and pers. obs.). The mosaic of vegetation units within and surrounding the areas (measured from a vegetation map) was variable but

not different between both treatments, indicating that they do not differ either in terms of long-term disturbance history or in the general physiographic characteristics (Cingolani et al., 2004).

To estimate present stocking rate of each area we measured frequency of cattle, horse, sheep and goat dung (following Cingolani et al., 2003) in 250 randomly placed 30 cm × 30 cm squares in September 2003. Additionally, during 10 months (from September 2003 to the end of June 2004) we visited each area 11–21 times at regular time intervals and counted the livestock present.

To determine an index of *P. australis* browsing intensity for each area, within each of the 12 areas we selected 10 individuals that were 10–150 cm tall and at least partially accessible to livestock. We did not select individuals randomly because we wanted to obtain a browsing estimator independent of plant height (i.e. measured on individuals of similar average height in all areas). To achieve this goal, we divided the 10–150 range in three classes (10–50, 50–90 and 90–150 cm) selecting individuals within each class as evenly as possible. To select a higher total number of individuals, or individuals within a narrower range of heights was not possible due to the low density of *P. australis* in the areas. From September to November 2003, we counted the number of browsed and non-browsed stems for each individual and used the percentage of browsed stems as an estimator of browsing intensity (Palmer and Truscott, 2003). For the largest individuals we only measured two to four representative branches. Additionally, for each selected individual, we measured height, two perpendicular canopy widths and basal diameter (when many basal stems, the largest) as indicators of size, and number of basal stems as an indicator of shrubbiness (Renison et al., 2005). We also measured the surrounding environment of the selected individuals: the percentage of rock (1 m² area); the accessibility to livestock, with a code from 1 (low accessibility) to 4 (high accessibility). All these measures were performed to corroborate that mean selected individual height did not differ between treatments, and to test if treatments differ in other plant attributes or the surrounding environment of the selected individuals, to take this into account when interpreting results.

To explore what factors affect the degree of *P. australis* browsing within an area, for three of the

selected areas (two with reduced and one with high grazing pressure) we measured the same variables as above for 14 additional individuals (totalling 24 individuals per area), widening the range of individual heights (from 4 to 250 cm).

To characterize the 12 areas in terms of physical variables and corroborate whether there were no differences between treatments that could confound the interpretation of results, we randomly located 6–11 plots of 10 m × 10 m in each area and measured the following topographic characteristics: altitude above sea level, topographic position (in categories from 1 to 5, from valley bottoms to convex summits), slope inclination (%), slope aspect (degrees from the north) and roughness (measured as the difference between the maximum and the minimum height in the plot). Additionally, we measured the proportion (%) of natural rock outcrops and exposed rock due to soil erosion (to discriminate between both types of rock, we followed criteria established in Cingolani et al., 2003, 2004). We did not measure soil characteristics but previous studies indicate that soil properties are strongly associated to topography (Cingolani et al., 2003; Enrico et al., 2004).

To characterise the 12 areas in terms of biological variables, in the same 10 m × 10 m plots we visually estimated the cover percentage of different categories of plant types: *P. australis*, other woody species (mid sized and dwarf shrubs), tussock grasses with thin leaves (*Deyeuxia hieronymi*, *Festuca tucumanica*, *F. hieronymi* and others of less abundance), tussock grasses with broad leaves (*Poa stuckertii*), short perennial graminoids, annual graminoids, cactus, ferns, short forbs, tall forbs and mosses plus lichens. We also estimated total plant cover of non-rocky surface (i.e. discounting bare soil).

To estimate *P. australis* size structure, we measured the number, height and two perpendicular canopy widths of all *P. australis* individuals present in the same 10 m × 10 m randomly placed plots. Note that the plots do not necessarily include the individuals selected to measure browsing.

2.2.2. Data analyses

We calculated stocking rates for each area by averaging all the livestock counts per area performed during the 10 months. Since observations occasionally involved other animals than cattle we transformed all

observations into Cattle Equivalents (CE, Cocimano et al., 1977). To estimate the effective grazing pressure on vegetation, we relativised stocking rate, as well as dung frequency to the vegetated area, by discounting rock surface.

Variables measured in the 10 selected *P. australis* individuals and their surrounding environment (browsing percentage, height, basal diameter, surrounding rock percentage and accessibility) were averaged per area (not considering the 14 additional individuals measured for 3 areas), to obtain only one value for each area. We did not consider canopy width because this variable (estimated as the square root of the product between both width measures) was highly correlated with height for all *P. australis* individuals measured in this study ($R = 0.92$, $P < 0.001$, $N = 990$, including all selected individuals in the areas plus individuals measured in the 10 m $\times$ 10 m plots). To have an indicator of browsing variability within each area, we also calculated the coefficient of variation of browsing percentage between individuals.

The physical and biological variables measured in the 10 m $\times$ 10 m plots were also averaged, obtaining one value for each area. For slope aspect, the variable was decomposed in two before averaging: relative north and relative east aspect, by cosine and sin transformations, respectively, multiplied by slope inclination. These variables ranged from highly negative values (for sites with south or west aspect, for the first and the second variable, respectively, with steep slopes) to highly positive values (for sites with north or east aspect with steep slopes). Zero values represent flat sites, or sites with east or west aspect for the first variable, and north or south aspect for the second (Cingolani et al., 2002).

From the data measured in the 10 m $\times$ 10 m plots we also calculated total density of *P. australis* (individuals/ha) as well as average height, obtaining one value per area. Total density was calculated for each area by dividing the total number of individuals counted in all 10 $\times$ 10 plots by the total area summed by all plots. The same total number of individuals was considered to calculate average height per area. Additionally, to estimate the size structure for each area, individuals were categorised in size classes, and we calculated the density for each size class (from 4 to 30, 31 to 60, 61 to 90, 91 to 120, 121 to 150, 151 to 200, >200 cm), in the same way as for total density.

Canopy width was not considered for the reasons explained above.

We compared the values of all variables between both treatments (high and reduced grazing intensity). Variables compared included those obtained from selected individuals and their surrounding environment, physical and biological variables, mean *P. australis* height, total *P. australis* density and density per size class. The comparisons were performed with *t*-tests for independent variables, with equal or unequal variance according to the case.

Additionally, we analysed within-area variability of browsing in relation to other variables for the three areas where 24 individuals were measured. To achieve this, we performed Spearman rank correlations between browsing percentage and the remaining five variables measured on selected individuals (height, number of basal stems, basal diameter, accessibility index and total rock cover surrounding the individual) for the three sites separately and for all 72 individuals pooled.

2.3. Enclosure experiment

The experiment was performed in the northern portion of the Sierras Grandes range, at 2270 m a.s.l. To analyse whether browsing increases *P. australis* seedling mortality or retards their growth, in late Spring 1998 (December) we transplanted 25 seedlings (7 months age) to a recently constructed enclosure of 10 ha, and 25 seedlings to a heavily grazed adjacent area (stocking rate ca. 3 CE/ha) with similar physical and biological characteristics. Mean height of transplanted seedlings was 3.88 (0.5–12 cm) and seedlings were assigned randomly to both treatments. All seedlings were watered immediately after transplant, but never again. We recorded survival and height each winter (July or August), during 6 years. Additionally, in the last winter (2004), we measured browsing percentage as in the observational study.

We compared differences in survival between treatments with χ^2 test, and the differences in height and browsing of 6-year-old seedlings with a *t*-test for independent samples. In this last case, only individuals that survived till the 6th year could be compared. Thus, to corroborate that there were no initial differences in height among individuals that survived

Table 1

Mean values and ranges (between brackets) of grazing pressure indicators in the treatments with high and reduced grazing intensity

	Grazing intensity		<i>P</i> -values
	High (<i>N</i> = 7)	Reduced (<i>N</i> = 5)	
Effective dung frequency (%) ^a	20.5 (10.2–32.8)	7.1 (5.5–9.6)	<0.01
Effective stocking rate (CE/ha) ^b	2.24 (1.1–4.8)	0.16 (0–0.43)	<0.01

P-values from their respective *t*-tests are indicated.^a Effective dung frequency = dung frequency × 100/(100 – total rock cover).^b Effective stocking rate (Cattle Equivalents/ha) = stocking rate × 100/(100 – total rock cover).

in both treatments we performed a *t*-test comparing initial height of surviving individuals.

3. Results

3.1. Observational study

As expected, treatments with high and reduced grazing intensity significantly differed in effective dung frequency and stocking rates. The range of values obtained for each treatment confirmed that all areas included in management units where animal numbers were reduced, effectively had less in situ stocking rates than all areas under traditional management (Table 1). Average browsing on selected *P. australis* individuals was very high (89%) in heavy grazed areas, and moderate (48%) in areas with reduced grazing intensity (Table 2). Additionally, areas with heavy grazing had less variability of browsing pressure between individuals, as indicated by the marginally significant difference in the variation coefficients. As was intended when selection was performed, neither the average height nor other

individual *P. australis* characteristics differed between treatments (Table 2). In contrast, the surrounding environment of selected individuals was different between treatments. In areas with high grazing intensity, *P. australis* individuals were significantly less accessible, and surrounded by more rock than individuals in areas with reduced grazing (Table 2).

The physical characteristics of the areas (altitude, topographic position, roughness, slope inclination, relative north and east aspect, sun trajectory, natural rock outcrops and rock exposed by erosion) did not differ between treatments (*P* > 0.05). In contrast, some biological variables showed differences. Perennial graminoids were more abundant in areas with high grazing pressure, while tussock grasses with thin leaves, and mosses and lichens, were more abundant in areas with reduced grazing intensity (Table 3).

P. australis total density and individuals' mean height (measured in the 10 m × 10 m plots) did not differ between treatments. Total *P. australis* density was 919 and 643 individuals/ha (*P* = 0.28), and mean height 64 and 84.1 cm (*P* = 0.18) for areas with high and reduced grazing intensity, respectively. However, when considering size structure, density of individuals

Table 2

Mean values and standard errors of selected *P. australis* characteristics and their surrounding environment in treatments with high and reduced grazing intensity

	Grazing intensity		<i>P</i> -values
	High (<i>N</i> = 7)	Reduced (<i>N</i> = 5)	
Browsing percentage	88.9 ± 1.4	48.4 ± 9.3	<0.001
Variation coefficient of browsing (%)	9.4 ± 1.6	49.2 ± 16.3	0.071
Height (cm)	57.8 ± 5.6	69.2 ± 5.14	0.183
No. of stems	3.21 ± 0.23	3.98 ± 1.10	0.532
Basal diameter (cm)	4.78 ± 0.89	4.91 ± 1.47	0.940
Accessibility	2.2 ± 0.14	3.0 ± 0.17	0.005
Total rock surrounding the individual (%)	41.4 ± 4.2	23.2 ± 1.8	0.007

P-values from their respective *t*-tests are indicated.

Table 3

Mean values and standard errors of biological variables (% cover) in treatments with high and reduced grazing intensity

	Grazing intensity		P-values
	High (N = 7)	Reduced (N = 5)	
<i>Polylepis australis</i>	3.6 ± 1.1	6.2 ± 0.9	0.115
Other woody species	7.7 ± 2.9	6.1 ± 1.3	0.669
Tussock grasses (thin leaves)	16.4 ± 3.5	29.5 ± 3.4	0.027
Tussock grasses (thick leaves)	9.1 ± 3.7	5.2 ± 2.6	0.441
Short perennial graminoids	15.5 ± 5.0	8.7 ± 3.5	0.025
Annual graminoids	0.15 ± 0.12	0.05 ± 0.04	0.093
Cactus	0.14 ± 0.12	0.06 ± 0.06	0.226
Ferns	2.7 ± 1.1	5.5 ± 2.2	0.100
Short forbs	19.3 ± 9.0	11.4 ± 2.6	0.116
Tall forbs	4.1 ± 2.6	3.5 ± 1.7	0.665
Mosses and lichens	2.1 ± 0.3	3.6 ± 0.5	0.034
Total cover of non-rocky surface	96.3 ± 0.9	95.8 ± 1.0	0.720

P-values from their respective *t*-tests are indicated.

differed between treatments depending on the height class (Fig. 1). Density of individuals lower than 60 cm as well as taller than 200 cm did not differ between both treatments while density of individuals between 60 and 200 cm height was significantly higher in the areas with reduced grazing intensity (only marginally significant for 60–90 cm, Fig. 1).

For the three areas where browsing percentage was measured in 24 individuals the within-area correla-

tions showed that browsing percentage was negatively correlated with height and basal diameter in one of the areas with reduced grazing (the area with the highest height range of selected individuals), and for the whole data pooled (Table 4 and Fig. 2). Additionally, browsing percentage was positively correlated with accessibility for all the three areas independently, but not for the whole data pooled (Table 4 and Fig. 3).

3.2. Experiment

Seedlings of *P. australis* planted in heavy grazed and non-grazed conditions differed in survival and growth. Survival after 6 years was significantly higher in the ungrazed treatment than in the grazed treatment (60% and 28%, respectively; $P = 0.02$). Mortality occurred mainly in the 1st year (Fig. 4a). In the ungrazed treatment it continued until the 3rd year, while in the grazed treatment, mortality continued until the 5th year after planting (Fig. 4a). Seedlings that survived until the end of the experiment did not differ in their original height among treatments (4.5 and 3.21 cm for the ungrazed and grazed treatments, respectively; $P = 0.24$). In terms of net growth, the 6-year-old seedlings were significantly taller in the ungrazed treatment than in the grazed treatment (50.4 and 12.6 cm, implying an annual average growth of 8.4 and 2.1 cm, respectively; $P < 0.01$, Fig. 4b). Browsing percentage was significantly higher in the grazed treatment than in the ungrazed treatment (92.4% and 0.03%, respectively; $P < 0.001$).

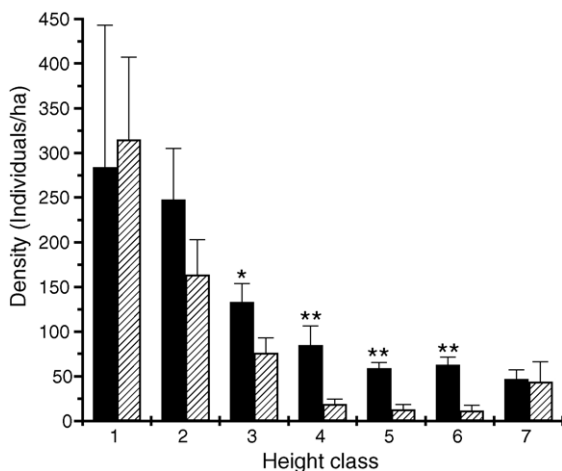


Fig. 1. *P. australis* average density and standard error for each height class: (1) <30 cm, (2) 31–60 cm, (3) 60–90 cm, (4) 90–120 cm, (5) 120–150 cm, (6) 150–200 cm, (7) >200 cm, in treatments with high (hatched) and reduced (solid) grazing intensity. Asterisks indicate significant differences between treatments: * $P < 0.1$; ** $P < 0.05$.

Table 4
Within-area analyses of *P. australis* browsing variability

	Area 1	Area 2	Area 3	All
Stocking rate (CE/ha)	0.12 (<i>N</i> = 24)	0.16 (<i>N</i> = 24)	4.80 (<i>N</i> = 24)	(<i>N</i> = 72)
Individual height	−0.53**	−0.26	−0.21	−0.64**
No. of stems	−0.30	0.12	−0.01	0.10
Diameter	−0.46**	−0.11	0.20	−0.45**
Accessibility	0.51**	0.38*	0.40**	−0.09
Rock (%)	−0.22	−0.25	−0.07	0.16

For the three selected areas where 24 individuals were measured, we indicated the Spearman rank correlations of browsing percentage with other variables (characteristics of individuals and their surrounding environment). In the last column, the correlations for all the areas pooled (*N* = 72 individuals).

* *P* < 0.1.

** *P* < 0.05.

4. Discussion

Results from the observational study suggest that livestock, in areas with high stocking rates, is retarding or preventing the natural recovery of *P. australis* woodlands. Our experimental data planting *P. australis* seedlings in and out of an enclosure strongly supports this interpretation. These data evidenced that domestic herbivores can have a great impact on height-growth and survival of this species, as occurs in many other tree species (Zamora et al., 2001; Gómez et al., 2003; Cierjacks and Hensen, 2004).

Our browsing rate measurements showed that stocking rate strongly influences the browsing pressure suffered by *P. australis* individuals. In areas with heavy grazing, all measured individuals had most of their stems browsed, while in areas with reduced grazing

pressure, *P. australis* individuals were less browsed, and variability between individuals was quite higher. These results suggest livestock is not highly selective of *P. australis*, and individuals may escape browsing to a certain extent in areas with reduced stocking rate. This is supported by the within-area analyses, which showed that within the same area, the least accessible individuals are less browsed, but considering all areas together, browsing damage is not related with accessibility. This is owed to the clearly higher browsing damage in the areas with heavy grazing, even for individuals in nearly inaccessible positions (see Fig. 3). All these results suggest that *P. australis* does not have strong and effective physical or chemical defenses to evade browsing, as is common in other woody species (Bryant et al., 1991; Harborne, 1997). On the contrary, the high browsing rate experimented

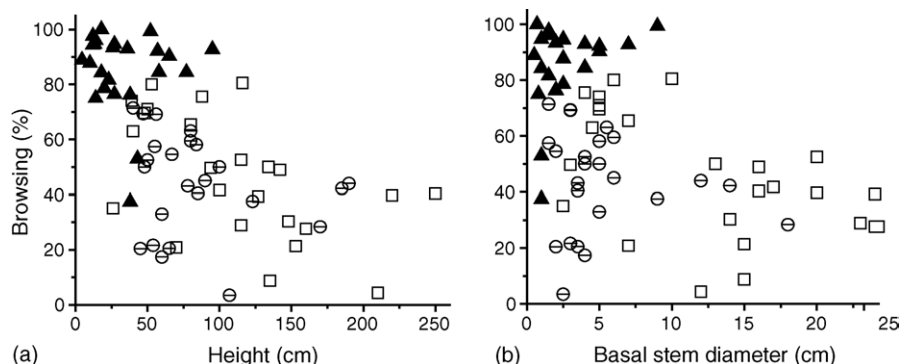


Fig. 2. Relationship of *P. australis* browsing percentage with (a) height and (b) basal stem diameter, for three areas with 24 individuals each, with different grazing intensities: (□) area 1, with reduced grazing intensity (0.12 CE/ha), (○) area 2, with reduced grazing intensity (0.16 CE/ha) and (▲) area 3, with high grazing intensity (4.8 CE/ha).

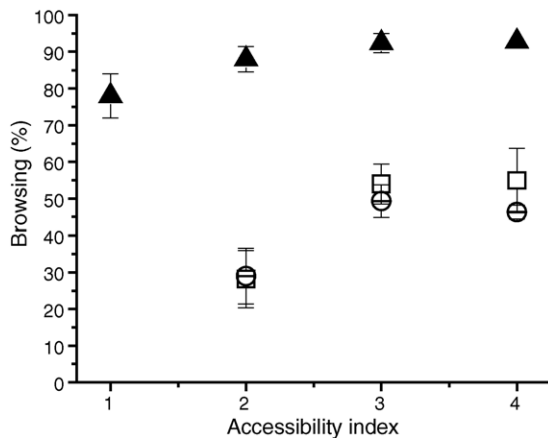


Fig. 3. Relationship between *P. australis* browsing percentage and accessibility index for three areas with 24 individuals each. We plotted mean browsing percentage for each value of the accessibility index, with the standard error: (□) area 1, with reduced grazing intensity (0.12 CE/ha), (●) area 2, with reduced grazing intensity (0.16 CE/ha) and (▲) area 3, with high grazing intensity (4.8 CE/ha).

by this species, and its notorious regrowth capacity (Renison et al., 2002, and pers. obs.), together with a relatively high nitrogen content (Vendramini et al., 2000, 2002) suggest that this species has a tolerant strategy to cope with herbivory (sensu Rosenthal and Kotanen, 1994). Considering that most nutritious forage grasses became almost unavailable in winter due to frosts and drought, animals increase the consumption of *P. australis* as the cold and dry season advances (Teich et al., unpublished data). Frequently, woody species are an alternative forage resource in winter, especially in highly seasonal systems (Posse et al., 1996; Rao et al., 2003).

Besides the differences in browsing damage, we found differences in the size structure between both treatments. These differences were similar to the findings of Motta (2003) in the Italian Alps for sensitive Norway spruce (*Picea abies*), after an increment in ungulate impact. In our case, we assume that the long-term history of disturbance was not different between treatments. Hence, the transition rates between size classes and thus the size structure was similar before livestock reduction. This assumption is supported by the lack of differences between groups in terms of the proportion of vegetation units, the percentage of exposed rock due to erosion, and adult *P. australis* density (i.e. individuals >200 cm), all variables that would be clearly different if long-term disturbance were different (Cabido and Acosta, 1985; Renison et al., 2002, in press; Cingolani et al., 2003, 2004). Additionally, both treatments showed the typical reverse J-shape that suggests a relatively stable size structure, with a low incidence of large disturbances (such as fires or windstorms; Frelich, 2002). Physical characteristics were not different between treatments, and most biological variables were also similar, except those expected to change rapidly with changes in the grazing regime (Pucheta et al., 1998; Cingolani et al., 2003), further supporting the assumption that present differences in size structure were caused by the recent reduction of grazing.

The density in the different size classes is the result of seed rain, seed viability, seed germination, seedling survival and height-growth (Schupp, 1995; Hulme, 1996). The fact that we determined higher densities of *P. australis* in size classes from 60 to 200 cm suggests

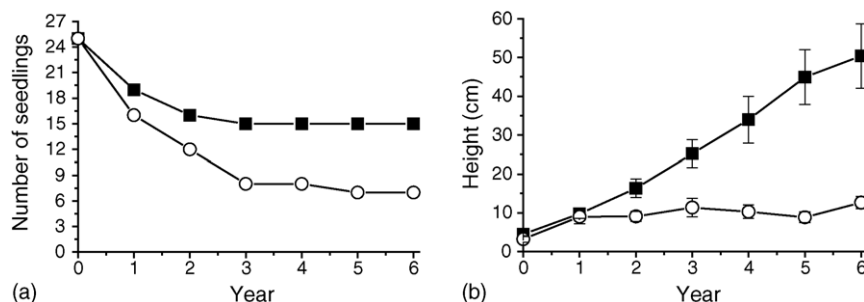


Fig. 4. (a) Number of surviving *P. australis* seedlings as a function of time (years) after planting. (b) Height of surviving individuals and standard error as a function of time (years) after planting. Filled squares (■) indicate the ungrazed treatment, and empty circles (○) indicate heavy grazed treatment.

that the reduction in grazing pressure influenced at least one of these life cycle parameters. Based on the experimental results and previous studies (Renison et al., 2004; Torres, 2004), we consider that the parameters most affected were seedling survival and height-growth of the size classes most accessible to livestock.

The experiment demonstrated that seedling survival and height-growth are strongly affected by heavy browsing. These results suggest that in the less browsed areas of the observational study, after livestock were reduced, seedlings and saplings began to grow more quickly and survived longer than in areas with high livestock stocking rates. In contrast, seed rain and seed viability are probably not different between treatments. In the case of seed rain, this is suggested by the fact that adult tree density, and thus the number of fruit bearing trees, did not differ between treatments. In the case of seed viability, known predictors of this parameter, such as the proportion of exposed rock due to erosion (Renison et al., 2004), were not different between treatments. Seed germination, in contrast, can be slightly influenced by stocking rate. According to previous studies (Torres, 2004), the presence of tussock grasses influences negatively on the germination of *P. australis* (indirectly measured as the presence of seedlings <5 cm). Thus, grazing reduction, by increasing the proportion of tussocks, can indirectly reduce seed germination, as known for other tree species (Prach et al., 1996; Rooney and Waller, 2003). However, in our study the differences in tussock cover between treatments were relatively small, and other parameters known to influence seed germination, such as exposed rock due to erosion, were not different between both treatments. This suggests that the influence of present grazing on seed germination is not very high, compared with the stronger effects on survival and growth.

The higher height-growth of areas with reduced grazing is reflected in the higher density of classes taller than 60 cm, but not in smaller classes. As suggested by the experimental results, mortality might be higher under heavy grazing, and probably occurred when individuals were small. This was evidenced by the lower accessibility of selected individuals (those used to measure browsing) in heavy grazed areas, which suggest that more accessible individuals are

extremely browsed and thus, less able to survive. The higher mortality in heavy grazed areas would produce a low density of small individuals, but this seems to be compensated by a slightly higher germination and a quite slower height-growth, that produced an accumulation in the smallest size classes of individuals of different ages (i.e. individuals remain in the small size class during longer periods than in areas with reduced grazing).

Summarizing, our results showed that in only 6–7 years, a reduction of grazing can produce significant changes in size structure, although these changes were not reflected in an increase in total *P. australis* density. However, our results suggest that if the present management is maintained, adults will increase their density in areas with reduced stocking rates, due to the height-growth of individuals 60–150 cm. Eventually, the increase in adult density will determine an increase of all size classes due to larger seed production.

4.1. Management implications

In the experiment, *P. australis* seedlings excluded from grazing grew on average 8 cm per year, which is comparable to the growth in other study areas under similar conditions (Renison et al., 2002, 2005), and to the annual growth rate known from other *Polylepis* seedlings (e.g., *P. besseri*, Hensen, 1994). In contrast, seedlings that survived in the heavily grazed adjacent area grew on average only 2 cm per year. This means that if the differences in growth rate between treatments are maintained, it will take browsed seedlings approximately four times longer to reach a height that allows escape from heavy livestock damage. These results indicate that protection from grazing can greatly accelerate the natural and assisted recovery of woodlands, and it would be the best option in cases where this objective is the only priority, such as in some parts of the National Park.

Data of our observational study suggests that a reduction in stocking rate does reduce browsing impact, although woodland recovery would probably be slower than in the case of complete exclusion. This would be an acceptable option for cases where livestock is necessary due to conservation or productive reasons. However, our results clearly indicate that the conservation and regeneration of *P. australis* woodlands requires the prevention of the

negative impacts produced by heavy stocking rates, which are a common practice in the study area. To promote woodland restoration in sites where livestock is necessary (such as in some sectors of the National Park), we therefore suggest a reduction in stocking rates, at least during time-lapses large enough for small individuals to reach sufficient height to escape livestock browsing.

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